

Instrumentation of the Prosthesis PrHand Based on Soft-Robotics: Angle sensor with Optical Fiber

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Abstract: The development of a fiber optic sensor based on intensity variation for angle measurement of the PIP joint in the hand prosthesis PrHand based on soft-robotics is presented and discussed its viability

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1. Introduction

It is estimated that by 2050 there will be 3.6 million amputees in the United States [1]. In 2018 around 59,000 amputations were performed in Brazil [2] and in 2019 it was estimated that there were more than 528,000 people with disabilities in their hands and legs [3] in Colombia. In 2011, it was estimated that in developing countries, around 30 million people with amputations did not have any assistive devices, which is one reason for the increased interest in developing robotic hands to help improve the quality of life of these people [1]. Robotic hands are intended to help improve users' self-esteem, and psychological problems and support activities of daily living [4].

PrHand is a hand prosthesis, described in [5], which is an electro-pneumatic device based on soft robotics because it has elastic joints that make the extension of the fingers; the flexion is done with tendons driven by a servomotor. Also, the fingers have the degrees of freedom (DOF) of abduction, and adduction that are done with air-controlled silicone actuators. The fingers are based on a compliant mechanism that allows them to make flexion, extension, abduction, and adduction. The prosthesis is underactuated because it can control 15 DOF with one motor and one pump- air. The control is performed in ROS on a Raspberry Pi 3. Figure 1.

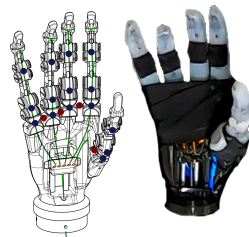


Fig. 1. PrHand prosthesis based on soft-robotics.

One of soft robotics devices main problems is the commercial sensors because most of them are made with rigid and industrial materials [6]. For that reason, optical fiber is taking place in the world of soft robotics, because one of its principal advantages is its flexibility. This paper reports the development of a sensor with optical fiber for the PrHand prosthesis. The proposed sensor monitors the angle of the proximal interphalangeal joint of the fingers.

2. Angle sensor

A polymeric optical fiber (POF) was used to develop the sensor, due to its many advantages including flexibility, low cost, resistance to electromagnetic interference, low weight, impact resistance, and high deformation. The working principle of the sensor is based on intensity variation, which means that it is measured variations in the output of the optical fiber considering a variation in the curvature of the fiber. For having a good variation of optical attenuation, a lateral cut is made that creates a sensitivity zone where there are power losses. The total

length of the fiber, the size of the cut and the depth will be factors that will influence the power variation [7]. To measure the angle in the proximal interphalangeal joint of the fingers, it is made a cut in the fiber for that point and then located in the finger.

3. Benchmark

Six cycles of opening and closing the hand were performed for the evaluation. Every 20 seconds the angle of the motor was changed by approximately 30° until the hand closed or opened completely, and the measurement of the voltage in the photodetector was made with a multimeter. For the measurement of the angle, the Kinovea software was used, where it is possible to track points, for that three points were placed as markers on the finger of the prosthesis, for the taking of power, 100 samples of the voltage variation were taken during the 20 seconds between change and change, and an average was made.

4. Preliminary results and discussion

Figure 2 is the graph of the response of the little finger sensor. This shows that the finger is handled between the ranges of plus or minus 90° to 20° , and the R^2 showed that a linear relationship between voltage and angle. In the first part of the graph, the points are concentrated at 90° because the way the fingers close the first angles are very close.

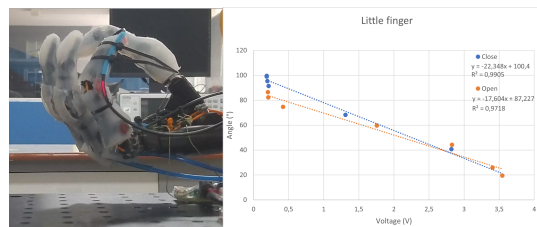


Fig. 2. Response of the angle sensor.

Considering the prosthesis is based on soft-robotics, the fiber is an alternative solution to measure parameters that allow controlling the behaviour of the prosthesis. The advantages it has regarding resistance to electromagnetic interference, flexibility, and cost, are significantly in this type of development, especially because one of the main goals of this prosthesis is low cost. The sensors are quite easy to implement and allow for replication, which is an essential feature of this type of development. One of the planned improvements is to enhance the way data is collected, since the system is calibrated as easily and quickly as possible. In general, all the fibers showed a good variety of voltage (in mean 2.4 Vcc) concerning the angle, which shows that the hand angles can be tracked during the movement.

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